Seat No :________ **No. of Printed Pages : 02**

[46]

Sardar Patel University
S.Y.BCA (SEM-3)(CBCS) Examination Feb-2020(On Demand)
10:00 a.m. to 01.00 p.m.
US03CBCA25 : Data Structure - I

12/02/2020 **Maximum Marks : 70**

Note: - Answers of all the questions (including multiple choice questions) should be written in the provided answer book only.

Q-1 Select the correct answer for the followings: **[10]**

1. ________ are the building blocks of Program?
   A. Data Structure B. Information C. DATA D. None.
2. ___________ Array is also known as matrix.
   A. One Dimensional B. Two Dimensional C. Both D. None.
3. Index is also known as _____.
   A. Sub Script B. None Sub Script C. Base D. Type.
4. A stack is _____ type of data structure.
   A. Linear B. Non Linear C. Both A and B D. None.
5. The term Push and Pop is related to _____________.
   A. Array B. Stack C. Queue D. None.
6. Prefix notation is called ______________.
   A. Suffix Notation B. Reverse Notation
   C. Polish notation D. None
7. A Queue is __________ Data structure.
   A. Linear B. Non primitive
   C. Homogeneous D. All of above
8. The linear formed for paying fees for college is example of _____?
   A. Stack B. Simple Queue C. Circular Queue D. None
9. ________ is meaningful information about entity.
   A. Record B. Entity C. Item D. None
10. A directed edge is known as _____________.
    A. Segment B. Arc
    C. Double edge D. Arrow.

Q-2 Write answers for the followings: (*ANY TEN*) **[20]**

1. Define: **Data Structure**.
2. List applications of **Data Structure**.
3. Define **linear data structure**. List out linear data structure.
4. List types of **Notation**.
5. Explain **Prefix** notation.
6. Define **Stack**. List applications of **Stack**.
7. What are Circular Queues?
8. State the limitation of **Simple Queue**.
9. What is **Queue**? List types of **Queue**.
10. Define **Complete Graph, Connected Graph**.
11. What is **Serial Processing**?
12. What is **Buffer**? Why buffers are used in sequential file organization.

Q-3 [A] Explain classification of **Data Structure**. [5]
[B] Explain representation of **1D array** in memory. [5]

OR

Q-3 [A] Explain representation of **2D array** in memory. [5]
[B] Explain applications of an **Array**. [5]

(P.T.O.)

Q-4	[A] Write an algorithm to Insert an element into a Stack.	[5]
	[B] Explain **Infix** to **Postfix** conversion.	[5]
	OR	
Q-4	[A] Write an algorithm to Delete an element from a Stack.	[5]
	[B] Give Postfix form for **(A+B)*C/D+E^F/G**.	[5]
Q-5	[A] Explain **Priority Queues** with example.	[5]
	[B] Write an algorithm to Delete an element from Circular Queue.	[5]
	OR	
Q-5	[A] Write an algorithm to Insert an element into Circular Queue.	[5]
	[B] Write an algorithm to Delete an element from Simple Queue.	[5]
Q-6	Write short note on **Multiple Buffering** and **Single Buffering**.	**[10]**
	OR	
Q-6	Explain in detail the **Structure of Sequential file** and **Processing of Sequential File.**	**[10]**